

ODE $\dot{x} = f(x)$, $x \in D \subseteq \mathbb{R}^n$

$x(t)$ & $y(t)$ are two solⁿ starting with diff IC

$$\delta x = y - x$$

$$\Rightarrow \dot{\delta x} = \frac{\partial f}{\partial x} \delta x$$

$$\boxed{\dot{\delta x} = A(x) \delta x} \quad \text{--- (1)}$$

Jacobian of f

Squared length of virtual disp

$$V = \delta x^T \delta x$$

$$\dot{V} = 2\delta x^T A(x) \delta x = \delta x^T (A(x) + A(x)^T) \delta x$$

If symmetric part of A satisfies

$$\frac{A(x) + A(x)^T}{2} \leq -\lambda I, \lambda > 0$$

$$\text{let } \lambda = -\gamma \therefore \gamma > 0 \Rightarrow -\gamma < 0$$

$$\Rightarrow \dot{V} \leq -2\lambda V \quad ||\delta x(t)|| \leq e^{-\lambda t} ||\delta x(0)||$$

Riemann Contraction theory

Let state dependent Riemann metric $M(x) > 0$

$$V(x, \delta x) = \delta x^T M(x) \delta x, M(x) = \Theta(x)^T \Theta(x)$$

Let $\delta z = \Theta \delta x \Leftarrow$ Transformed $\delta x \rightarrow \delta z$

$$\delta x = \Theta^{-1} \delta z \quad \boxed{V = \delta z^T \delta z = \delta x^T \Theta^T \Theta \delta x = \delta x^T M \delta x}$$

$$\dot{\delta z} = \dot{\Theta}(x) \delta x + \Theta(x) \dot{\delta x}$$

$$\dot{V} = \delta x^T \dot{M} \delta x + \delta x^T M \dot{\delta x} + \dot{\delta x}^T M \delta x = \delta x^T (\dot{M} + A^T M + M A) \delta x$$

metric change

$$\dot{S}z = (\dot{\Theta}(x) + \Theta(x)A(x)) \dot{\Theta}^{-1} Sx$$

where Lie derivative $\dot{\Theta}$ along f

$$\dot{\Theta}(x) = \frac{\partial \Theta}{\partial x} f(x) = \sum_{i=1}^n \frac{\partial \Theta}{\partial x_i} f_i(x)$$

$$\dot{V} \leq -2\lambda V \Rightarrow Sx^T \dot{S}x \leq -2\lambda Sx^T M Sx$$

Generalized Jacobian $\Rightarrow V \leq Sx^T (\dot{M} + A^T M + M A + 2\lambda M) Sx \leq 0$

$$\dot{S}z = (\dot{\Theta}(x) + \Theta(x)A(x)) \dot{\Theta}^{-1} Sx$$

$$K = -\lambda I$$

$$\dot{S}z = K Sx$$

$$(\dot{\Theta} + \Theta A) \dot{\Theta}^{-1} = -\lambda I \Rightarrow K + \lambda I = N \dot{\Theta}^{-1}$$

$$\dot{\Theta} + \Theta A = -\lambda \dot{\Theta}$$

$$K = -\lambda I$$

$$\Rightarrow K + \lambda I$$

$$N = (K + \lambda I) \dot{\Theta}$$

$$\Rightarrow N = (\dot{\Theta} + \Theta A + \lambda I) \dot{\Theta}^{-1} \dot{\Theta} = \dot{\Theta} + \Theta A + \lambda I$$

$$N = \dot{\Theta} + \Theta A + \lambda I$$

Transform M can be used by SOS

System contracting iff symmetric part of K is uniformly -ve definite

$$\frac{1}{2} (K + K^T) < 0 \quad \forall x \in D$$

For exponential convergence

$$\frac{1}{2} (K + K^T) \leq -\lambda I$$

is a box
 $\begin{cases} a \leq |x_1| \leq b \\ |x_2| \leq b \end{cases}$ State

$$K + K^T \leq -2\lambda I$$

$$\dot{\Theta} + \Theta A + \lambda I + \dot{\Theta}^T (\dot{\Theta} + \Theta A) + 2\lambda I \leq 0$$

$$K + K^T + 2\lambda I \leq 0$$

we finally obtain

$$S(x) = \dot{M}(x) + A(x)^T M(x) + M(x) A(x) + 2\lambda M(x) \leq 0$$

when $M = \Theta^T \Theta$

$$\dot{M}(x) = \sum_{i=1}^n \frac{\partial M}{\partial x_i} f_i(x)$$

Example

a) $\dot{x} = \begin{bmatrix} -2 & 0 \\ -1 & -3 \end{bmatrix} x$ | Choose $\Theta = I$ | $\dot{\Theta} = 0$
 $M = \Theta^T \Theta = I$

$$S = A^T M + M A + 2\lambda M$$

$$S = A^T + A + 2\lambda I$$

$$S = A^T + A + 2\lambda \leq 0$$

$$A^T + A \leq -2\lambda$$

$$\begin{bmatrix} -2 & 0 \\ -1 & -3 \end{bmatrix} + \begin{bmatrix} -2 & -1 \\ 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & -1 \\ -1 & -3 \end{bmatrix} \leq -2\lambda$$

b) $\dot{x}_1 = -x_1 + x_1 x_2$, $\dot{x}_2 = -x_2$

$$f(x) = \begin{bmatrix} -x_1 + x_1 x_2 \\ -x_2 \end{bmatrix}$$

$$A = \frac{\partial f}{\partial x} = \begin{bmatrix} -1 + x_2 & x_1 \\ 0 & -1 \end{bmatrix}$$

Choosing $\Theta = I$, $\dot{\Theta} = 0$, $\dot{\Theta} + \Theta \frac{\partial f}{\partial x} = 0 + I A = A$

$$\frac{K + K^T}{2} = \frac{A + A^T}{2} = \frac{1}{2} \begin{bmatrix} -2 + 2x_2 & x_1 \\ x_1 & -2 \end{bmatrix}$$

Also. $M = \Theta^T \Theta = I$, if $\Theta = I \Rightarrow$
 $\mu = 0$

So $A^T + A + 2\lambda I \leq 0$

$\Rightarrow \frac{1}{2}(A^T + A) \leq -\lambda$

same

$$\begin{bmatrix} -2 + 2x_2 + 2\lambda & x_1 \\ x_1 & -2 + 2\lambda \end{bmatrix}$$

let $\lambda = +0.1$

$$S = \begin{bmatrix} -1.8 + 2x_2 & x_1 \\ x_1 & -1.8 \end{bmatrix}$$

$$\begin{aligned} -1.8 + 2x_2 &\leq 0 \\ \Rightarrow x_2 &\leq 0.9 \end{aligned}$$

$$\det(S) = s_{11}(-1.8) - x_1^2 > 0$$

$$\begin{aligned} \Rightarrow x_1 &< (-1.8 + 2x_2)(-1.8) \\ &= 1.8(1.8 - 2x_2) \end{aligned}$$

$$D_{\max} \{ x_1, x_2 \mid x_2 \leq 0.9 \text{ and } x_1^2 < 1.8(1.8 - 2x_2) \}$$

when we used solver we got

$$M = \begin{bmatrix} 0.792 + 0.259x_2 & 0.120x_1 \\ 0.120x_1 & 0.957 \\ & + 0.043x_2 + 1.083x_2^2 \end{bmatrix}$$